**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 260 · RAM, ROM and virtual storage

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why are SSDs and HDDs physically different?

Q2. What purpose does processor cache serve?

Q3. Which is normally volatile: RAM or ROM?

Q4. Sorted array [2,6,10,14,18,22,26,30]. Find 22 with midpoint indexes and range updates.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

RAM — Volatile read/write working memory.

ROM — Non-volatile firmware/boot storage.

virtual storage — Remote network-accessible file storage.

virtual memory — OS technique using address translation and backing storage.

firmware — Low-level code retained without power.

## New learning

RAM versus ROM — RAM supports active execution and is volatile; ROM and firmware storage retain instructions without power.

Define virtual storage — A remote or cloud storage service keeps files on network-accessible systems, potentially available on multiple devices.

Distinguish virtual memory — OS virtual memory involves memory addressing/paging with secondary storage, taught again in OS memory management.

## Worked example

Why can unsaved editing work disappear after power loss while firmware remains?

1. Predict — Unsaved temporary program state depends on volatile RAM and can disappear without power.

2. Explain — Firmware persists in non-volatile storage, allowing startup instructions to remain available.

# We Do · Practise with guidance

A pupil opens saved coursework from home using a school online drive. State advantages and risks.

```
Remote file storage accessed over network.
The file may sync to several devices.
```

Predict

Virtual/cloud storage supports access from multiple locations and may offer remote copies.

Explain

Network outages, account security and mistakenly synced deletions are relevant risks.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

State typical use and whether RAM and ROM are volatile.

### Question 2

Explain two benefits and one drawback of network-based virtual storage for pupil coursework.

### Question 3

Distinguish virtual storage from virtual memory and explain one limitation of each.

## Exit Ticket · Answer before checking

Virtual cloud storage is another name for OS virtual memory.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. SSDs use semiconductor cells; HDDs use rotating magnetic platters.

Q2. Keeps some needed data/instructions in fast nearby memory.

Q3. RAM is volatile.

Q4. low0 high7 mid3→14; low4 high7 mid5→22, found at index 5.

### You Do pages · Independent questions

Q1. [4] RAM running program/data [1] volatile [1]; ROM/firmware low-level startup data [1] non-volatile [1].

Q2. [6] Multi-device access [2], remote copies/scalability [2], connectivity or privacy/security issue [2].

Q3. [6] Virtual storage is remotely hosted files [2], susceptible to outages/security concerns [1]; virtual memory extends usable address space through paging and secondary storage [2], heavy paging harms speed [1].

### Exit Ticket · Model answer

False. The concepts differ.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**



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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check my understanding